

# BETWEEN WHISPERS AND SCREAMS

## LOUDNESS STANDARD DEVIATION AS A PROXY FOR EXPLICIT CONTENT DETECTION IN ROMANCE FICTION

Svenja Guhr



### RESEARCH QUESTION AND HYPOTHESIS

**Q:** To what extent does loudness SD serve as a proxy for identifying scenes flagged under SEX & NUDITY and VIOLENCE & SCARINESS (Tsai et al., 2024), annotated on a ternary scale (*absent, weak, strong*)?

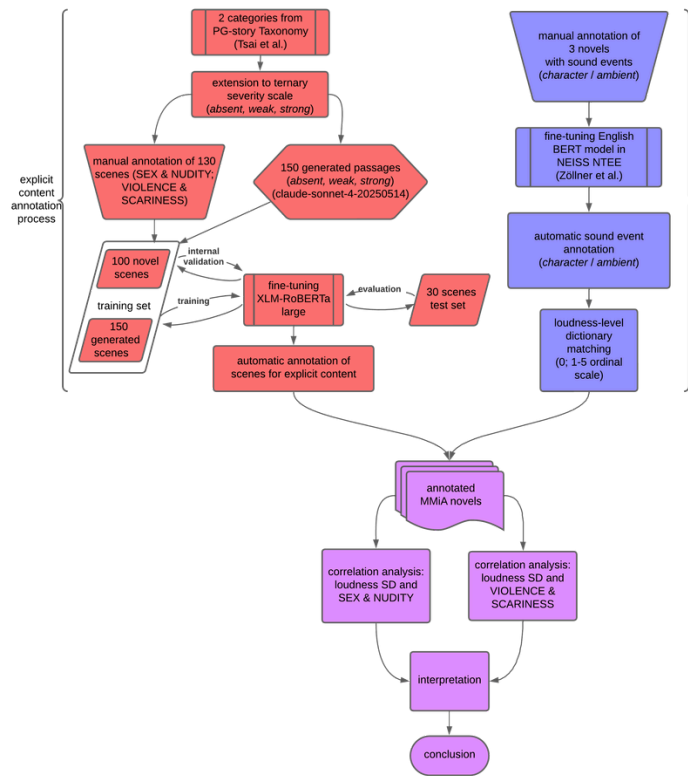
**H1:** Scenes annotated as *weak* or *strong* under VIOLENCE & SCARINESS will show significantly higher *ambient sound* loudness SD than *absent* scenes, reflecting the dynamic contrast between loud events and silence characteristic of threat.

**H2:** Scenes annotated as *weak* or *strong* under SEX & NUDITY will show significantly higher *character sound* loudness SD, reflecting the dynamic range from whispered dialogue to vocalized arousal.

### SOUND AND LOUDNESS IN MEN MADE IN AMERICA NOVELS

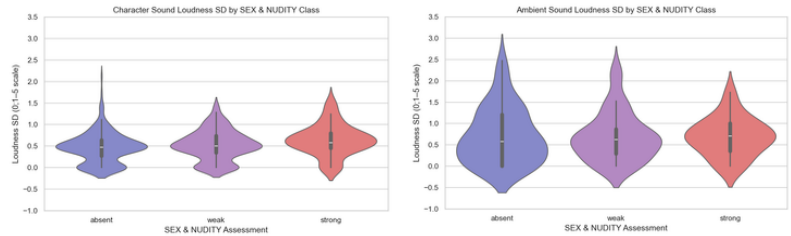
Guhr (2026) defines a fictional sound as a **sound event** with asserted *realis* in the fiction in line with the event concept by Gius/Vauth (2022), **explicitly** represented by **sound words** in the narrative. Sound events are classified into **ambient sounds** and **character sounds** and assigned **loudness levels on a 0; 1–5 ordinal scale** via dictionary matching.

**Loudness mean** and **Standard Deviation (SD)** are retained as corpus-comparative heuristics, interpreted as categorical spread enabling comparative analysis of sound profiles across texts.

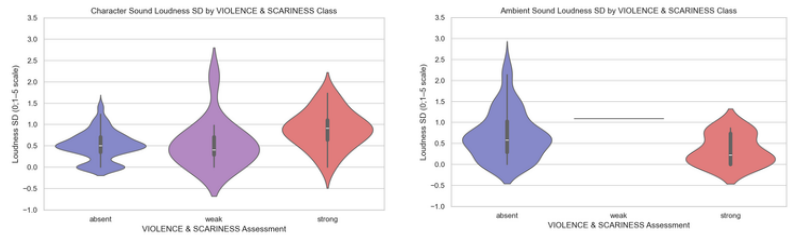


**Fig. 1:** Two parallel annotation pipelines (explicit content, left; sound & loudness, right) merging for two distinct correlation analyses of (1) loudness SD and SEX & NUDITY scenes, and (2) loudness SD and VIOLENCE & SCARINESS scenes.

### RESULTS



**Fig. 2:** Character sound loudness SD by SEX & NUDITY class. SD correlates positively with erotic scene score ( $p = 0.168$ ); mean SD rises from 0.467 (absent) to 0.644 (strong, +37.9%) (left). Ambient sound loudness SD by SEX & NUDITY class. No equivalent effect ( $p = 0.067$ ,  $p = .42$ ), ruling out general acoustic inflation (right).



**Fig. 3:** Character sound loudness SD by VIOLENCE & SCARINESS class. Violent/scary scenes show higher character SD ( $M = 0.749$  vs.  $0.509$ ,  $p = .004$ ) (left). Ambient sound loudness SD by VIOLENCE & SCARINESS class. Ambient SD returns null results across all tests ( $p = .661$ ); only 3 weak scenes have  $\geq 2$  ambient sound events, too few for meaningful comparison (right).

### CONCLUSION

Character sound loudness SD is an acoustically grounded proxy for explicit content in romance fiction.

**H2 confirmed:** SD correlates positively with erotic scene score ( $p = 0.168$ ,  $p < .001$ ), rising from 0.467 (absent) to 0.644 (strong, +37.9%;  $AUC = 0.658$ ,  $p < .001$ ). The SEX & NUDITY classifier achieved good results (macro  $F1 = 0.63$ ) even under low-resource conditions.

**H1 inconclusive:** Violent/scary scenes show higher character SD ( $M = 0.749$  vs.  $0.509$ ,  $p = .004$ ), but ambient SD returns null results ( $p = .661$ ), limited by sparse annotation ( $n = 14$ ).

Character sound SD is elevated in both categories, suggesting it may signal narrative intensity more broadly, not eroticism alone.

The VIOLENCE & SCARINESS classifier (macro  $F1 = 0.30$ ) proved harder to generalize, likely because the category bundles narratologically distinct phenomena. Future work will separate scariness from violence and use larger corpora.

### REFERENCES, CODE, AND DATA

Anthropic. 2025. Claude Sonnet (claude-sonnet-4-20250514). <https://www.anthropic.com>.  
Gius, Evelyn, and Michael Vauth. 2022. "Towards an Event Based Plot Model. A Computational Narratology Approach." *Journal of Computational Literary Studies* 1(1). 10.48694/jcls.110.  
Guhr, Svenja. 2026. Explicit Content Detection. GitHub Repository. [https://github.com/SvenjaGuhr/Explicit\\_Content\\_Detection](https://github.com/SvenjaGuhr/Explicit_Content_Detection).  
Guhr, Svenja. (in print). Raise Your Voice – Character Sound in German-language Fiction. J.B. Metzler Book Series "Digitale Literaturwissenschaft", Springer Nature.  
Tsai, Alicia Y., Shereen Oraby, Anjali Narayan-Chen, et al. 2024. "PG-Story: Taxonomy, Dataset, and Evaluation for Ensuring Child-Safe Content for Story Generation." *Proceedings of the Third Workshop on NLP for Positive Impact*, 78–97.  
Zöllner, Jochen, Konrad Sperfeld, Christoph Wick, and Roger Labahn. 2021. "Optimizing Small BERTs Trained for German NER." *Information* 12(11): 443. 10.3390/info12110443.